UNIT - 3

Capital structure theories & Cost of capital.

EBIT → Earning Before Interest tax

EBT → Earning Before Tax

EAT → Earning after tax

EPS → earning per share.

	₹
EBIT	
less : Interest	xxx
	xx
EBT	xxx
less : Tax	xx
EAT	xxx
no. of shares	xx

earning per share $EPS = \frac{EAT}{\text{no. of shares}}$

Q

	Plan I	Plan II
Debit (Interest @ 10% pa)	4,00,000	1,00,000
equity shares (₹ 10 each)	1,00,000	4,00,000
total finance required	5,00,000	5,00,000
No. of equity shares	10,000	40,000

EBIT = ₹ 50,000

Sol no. of shares $= \frac{1L}{10} = 10,000$

	Plan - I	Plan-II
EBIT	50,000	50,000
less: Interest	4,00,000×10%	100,000×10%
EBT	40,000	10,000
	10,000	40,000
less: Tax	5000	20,000
EAT	5000	20,000
no. of Shares	1000	40,000
Eps	₹ 1.75 per share	₹ 1.43 per share.

(11) A Company has Equity Share Capital of ₹ 5,00,000 divided into shares of ₹ 100 each. It wishes to raise further ₹ 3,00,000 for a Expansion ~~Cent~~ and modernization plants. The Company plans the following financing schemes.

a) All Common Stock

b) ₹ 1,00,000 in Common stock & ₹ 2,00,000 in 10% debentures.

c) All debt at 10% p.A.

d) ₹ 1,00,000 in common stock & ₹ 2,00,000 in preference capital with the rate of ~~discount~~ divident at 8%.

The Company's Expected EBIT ₹ 1,50,000, Coorperate rate of Tax is 50%. You are required to determine Eps & Comment

	Plan I	plan II	plan III	plan IV
equity	₹ 3,00,000	₹ 1,00,000	–	₹ 1,00,000
debentures	–	₹ 2,00,000	₹ 3,00,000	–
preference	–	–	–	₹ 2,00,000
	₹ 3,00,000	₹ 3,00,000	3,00,000	3,00,000

	plan - I.	Plan - II	plan-III	plan-IV
EBIT	1,50,000	1,50,000	1,50,000	1,50,000
less: Interest	–	20,000 [₹2L × 10%]	30,000 [₹3L × 10%]	–
EBT	1,50,000	1,30,000	1,20,000	1,50,000
less: Tax 50%	75,000	65,000	60,000	75,000
Earning after Tax →	75,000	65,000	60,000	75,000
less: preference dividend	–	–	–	16,000 [2L × 8%]
Earning available to equity Share holders	75,000	65,000	60,000	59,000

Existing share - [25L/100]		5000	5000	5000	5000
New share	30	3,000	1,000	–	1000
earnity ÷ 100		8,000	6,000	5,000	6,000

Earning per share = Earnings available to equity to share holder / no. of equity shares.

Plan - I	Plan - II	Plan - III	Plan - IV
75,000 / 8,000 sh	63,000 / 6,000 sh	60,000 / 5,000 sh	59,000 / 6,000 sh
= ₹9.375/sh	₹10.8/sh	₹ 12/sh	₹9.83/sh

∴ the Eps is higher in plan III

XYZ & external Company has currently an equity share capital of ₹40L consisting of 40,000 equity shares of ₹100 each. The management is planning to raise another ₹ 30L to finance a major programme of expansion through one of the four possible financing plans. debentures long term } Interest

The option are :-

i) Entirely through equity shares.

ii) ₹15L in equity shares of ₹100 each & the balance in 8% debentures

iii) ₹ 10L in equity shares of ₹100 each & the balance through long-term borrowing at 9% interest p.A. 30-10 = 20 debentu

iv) ₹15L in equity shares of ₹100 each & the balance through preference shares with 5% dividend.

The company's expected earning before interest & tax as (EBIT) will be ₹ 15L. Assuming corporate tax rate of 50%, you are required to comment on the financial coverage that will be authorized under each of the above scheme of financing.

Calculation of no of equity shares

	Plan I	plan II	Plan III	Plan IV
5) equity				
Existing shares no. of shares ₹ 40L/100	40,000	40,000	40,000	40,000
i) no. of equity shares ₹ 30L/100	30,000			
ii) Debentures eqush ₹ 15L/100		15L @ 8% 15000		
iii) eqush ₹ 10L/100		~~10,000 20L@9~~	10,000 20L@9%	
L.T Borrowing				15000 ₹15L@5%
iv) eq ₹ 15L/100				
preference share	70,000	55,000	50,000	35,000

Calculation of Eps.

	Plan - I	Plan - II	Plan - III	Plan - IV
EBIT	15,00,000	15,00,000	15,00,000	15,00,000
less: Interest	–	1,20,000 15L × 8%	1,80,000 20L × 9%	–
EBT →	15,00,000	13,80,000	13,20,000	15,00,000
Less: Tax 50% (-)	7,50,000	6,90,000	6,60,000	7,50,000
Earning after Tax →	7,50,000	6,90,000	6,60,000	7,50,000
Less: preference dividend	–	–	–	75,000
Earning available to equity shareholders →	7,50,000	6,90,000	6,60,000	6,75,000
Eps = / no. of equity shares	75,0000 / 30,000	6,90,000 / 55,000	6,60,000 / 50,000	6,75,000 / 35,000

=₹10.71/sh ₹12.54/sh ₹13.2/sh ₹12.27/sh

∴ The Eps is higher in plan III

Calculate Eps (Earning per share) of shy Ltd and Smart Ltd assuming (a) 20% before Tax rate of return on assets (b) 10% before tax rate of return on assets based on the following data.

	Shy Ltd	Smart Ltd
Assets	2,00,00,000 ×20% EBIT	2,00,00,000
Debt (12%)	–	1,00,00,000
Equity	2,00,00,000 (shares of ₹10 each)	1,00,00,000 (shares of ₹10 each

Assume a 50% Income tax in both cases. Also give your comments on the financial leverage

Calculation of EPS

	Shy Ltd		Smart Ltd	
	Plan-I	Plan-II	Plan-I	Plan-II
EBIT	40L	20L	40L	20L
less: Interest (–)	–	–	12L	12L
EBT →	40L	20L	28L	8L
less: Tax @50%	20L	10L	14L	4L
EAT	20L	10L	14L	4L
less: preference/divide	–	–	–	–
Earning available to earning share equity.	20L	10L	14L	4L
no. of equity shares	200L/10	200L/10	100L/10	100L/10
=	20L	20L	10	10

Eps = Earning available to equity to share holder / no. of equity shares

20/20	10/20	14L/10	4L/10
₹1/sh	₹0.5/sh	₹1.4/sh	₹0.4/sh.

∴ Eps is higher in plan – I of Smart Ltd.

Capital Structure.

Capital structure of a company refer to a composition or makeup of its capitalization ~~or~~ makeup of & it includes all long-term capital resources i.e (loans, reserves, shares & bonds).

Factors determining the capital structure.

(i) Financial leverage or trading or Equity
(ii) Growth & stability of sales.
(iii) Cost of capital
(iv) Risk
(v) Cash flow ability to service debt.
(vi) Nature & size of a firm.
(vii) flexibility
(viii) Control
(ix) Requirements of investors
(x) Capital market conditions
(xi) Assets structure
(xii) purpose of financing.

(xiii) period of finance
(xiv) cost of flotation
(xv) personal consideration
(xvi) Corporate Tax rate
(xvii) legal requirements

Theories of Capital structure.
(i) Net Income Approach : N
(ii) Net operating Income approach
(iii) Traditional Approach
(iv) Modigliani & miller approach.

(i) Net Income Approach (V = S+D)

Q) X ltd is expecting an annual EBIT of ₹ 1,00,000. The company has ₹ 4,00,000 in 10% debentures. The cost of Equity capital or Capitalization rate is 12.5%. You are required to calculate the total value of the firm according to the net income approach.

Sol V = Value of the firm
S = market value of equity
D = Debentures

V = S+D

Computation of value of the given firm.

Particulars	₹
EBIT	1,00,000
- Interest 4,00,000 @10%	40,000
EBT	60,000
- Tax	–
EAT	60,000
- Preference dividend	–
Earnings available to Share holders	60,000
Cost of equity capital or Capitalization	12.5%

Earnings available to sh holder / Cost of Equity $= \frac{60,000}{0.125}$ 4,80,000

V = S+D
V = 4,80,000 + 4,00,000
V = 8,80,000

Q) A firm is considering two financial plans with a view to examine they impact on EPS. Total funds required for investing in assets are ₹ 6,00,000

	Plan - I	Plan II
Debt Interest @ 12% PA	5,00,000	2,00,000
Equity shares @ ₹10 each	1,00,000 = $\frac{1,00,000}{10}$ = 10,000	4,00,000 = $\frac{4,00,000}{10}$ = 40,000

The EBIT is assumed as 75,000, 1,00,000, 1,25,000, Tax rate is 50%.

Particulars	Plan - I			Plan - II		
EBIT	75,000	1,00,000	1,25,000	75,000	1,00,000	1,25,000
less: Interest @ 12% Plan-I - 5L × 12% II - 2L × 12%	60,000	60,000	60,000	24,000	24,000	24,000
EBT	15,000	40,000	65,000	51,000	76,000	1,01,000
less: Tax @ 50%	7500	20,000	32,5000	25,500	38000	50,500
EAT	7500	20,000	32,500	25,500	38,000	50,500
- Preference dividend	–	–	–	–	–	–
Equity Shares	$\frac{1,00,000}{10}$	$\frac{1,00,000}{10}$	$\frac{1,00,000}{10}$	$\frac{4,00,000}{10}$	$\frac{4,00,000}{10}$	$\frac{4,00,000}{10}$
	10,000	10,000	10,000	40,000	40,000	40,000

$EPS = \frac{\text{Earnings available}}{\text{no. of shares}} = \frac{7500}{10,000}, \frac{20,000}{10,000}, \frac{32,500}{10,000}, \frac{25,500}{40,000}, \frac{38,000}{40,000}$

$= 0.75 \quad = 2 \quad = 3.25 \quad = 0.63 \quad = 0.95$

$= \frac{50,500}{40,000} = 1.26$

∴ Plan I - 3 is higher EPS

Net operating income Approach:-

Q A Company Expects a net operating Income of ₹ 1,00,000. It has 5,00,000 @ 6% debentures. The overall capitalization rate is 10%. Calculate the value of firm.

Sol = V = Value of the firm $\boxed{V = S + D}$

S = Value of Equity

D = Debentures.

Given = D = 5,00,000

overall capitalization rate = 10%

EBIT or net operating income = ₹1,00,000

$V = \frac{EBIT}{\text{overall Capitalization rate}} = \frac{1,00,000}{10\%} = \frac{1,00,000}{0.1} = 10,00,000$

V = S+D = 10,00,000 = S + 5,00,000.

S = 5,00,000 - 10,00,000, S = 5,00,000

Equity Capitalization rate = $\frac{EBIT - I}{\text{Total market value of Equity Shares}}$.

Rate of Equity.

= $\frac{\text{Earning available to Equity shares}}{\text{Shares}}$.

EBIT	-	1,00,000
less Interest	-	30,000
EBT	-	70,000
less. Tax	-	–
EAT	-	70,000
less Preference dividend	-	–

→ Earning Available = 70,000

$= \frac{70,000}{5,00,000}$

$= 0.14 \mid 14\%$

Q) There are two firms X & Y which are Exactly identical. Except that X does not use any debt in its financing while Y has 1,00,000 @ 5% debentures. both the firms have earnings before Interest & tax of ₹ 25,000 & the Equity capitalization rate is 10%. Assuming the Coorperation tax of 5%. Calculate the value of the firm using m & m

Sol. $S = \frac{EBIT}{\text{Cost of Equity}} (1 - t)$ → Tax rate

$= \frac{25000}{0.10} (1 - 0.5)$

$= 1,25,000$.

Cost of Capital

Cost of capital for a firm may be defined as the cost of obtaining funds i.e the average rate return that the investers in the firm. The cost of obtaining funds to expect for supplying funds to the firm

Significance of cost of capital

Classification of cost

1. historical cost & future cost
2. specific cost & composite cost
3. Explict cost & complict cost
4. Average cost & Marginal cost

Calculation of cost of capital

Calculation of overall cost of capital involves.

a) Calculation of cost of specific source of finance

b) Calculation of wrighted average cost of capital.

Calculations of cost of specific source of finance

1. Cost of debt, bonds & debentures
   - Irredeemable / Perpetual
   - redeemable

Irredeemable ~~Kdb~~ $K_{db} = \frac{I}{NP}$ Before Tax

$K_{db} = \frac{I}{NP}(1-t)$ After tax

I = Interest

Np = Net proceed

t = tax rate

Kdp = Cost of debt or bond

Reedeemable $K_{db} = \frac{I + \frac{RV-NP}{n}}{\frac{RV+NP}{2}}$ Before Tax

n = no. of years

Rv = redeemable value

$K_{db} = \frac{I(1-t) + \frac{RV-NP}{n}}{\frac{RV+NP}{2}}$ After tax

A Ltd issues 1,00,000 10% perpetual debenture of ₹100 each. Calculate the cost of net capital.

Sol $I = 1,00,000 \times 10\% = 10,000$

$K_{db} = \frac{I}{NP} = \frac{₹10,000}{1,00,000} = 0.1 = 10\%$

OR

$\frac{10}{100} = 0.1 = 10\%$

(i) at premium of 10%

$= \frac{10}{110}$; $NP = ₹100 + 10\%$ of 100

$= ₹100 + 10$

$= ₹110$

$= 0.0909$

$= 9.09\%$

(ii) at a discount of 5%

$NP = ₹100 - 5\%$ of ₹100

$= ₹100 - ₹5$

$= ₹95$

$\frac{₹10}{₹95} = 0.1052 = 10.52\%$

A Company issues 10,00,000 @10% redeemable debentures at discount of 5%. The cost of floatation amount ₹30,000. The debentures are redeemable after 5 yrs calculate before tax & after tax cost of debt assuming a tax rate of 50%.

Before tax $K_{db} = \frac{I + \frac{RV-NP}{n}}{\frac{RV+NP}{2}}$

here $I = ₹1,00,000$ 10,00,000 × 10%

NP ~~Rv~~ = 10,00,000 − 10,00,000 × 5% − 30,000

Total Amt − tot Amt × dis − floatation

10,00,000 − 50,000 − 30,000

= 9,20,000

Rv = ₹10L, n = 5 yrs

$K_{db} = \frac{1,00,000 + \frac{10,00,000 - 9,20,000}{5}}{\frac{10,00,000 + 9,20,000}{2}} = \frac{1,00,000 + \frac{80,000}{5}}{\frac{19,20,000}{2}}$

$1,00,000 + \frac{16,000}{9,60,000}$ = ~~1,00,000 + 0.0166~~

$= \frac{1,16,000}{9,60,000}$

$= 0.1208$ | 0.1208%

After tax $\dfrac{I(1-t) + \dfrac{RV-NP}{n}}{\dfrac{RV+NP}{2}}$

$= \dfrac{1,00,000(1-50\%) + \dfrac{10,00,000-9,20,000}{5}}{\dfrac{10,00,000+9,20,000}{2}}$

$= \dfrac{1,00,000(0.5) + \dfrac{80,000}{5}}{\dfrac{19,20,000}{2}}$

$= \dfrac{50,000 + \dfrac{80,000}{5}}{\dfrac{19,20,000}{2}}$

$= \dfrac{50,000 + 16000}{9,60,000}$

$= \dfrac{66,000}{9,60,000}$

$= 0.06875$

Q A 5 year ₹ 100 debentures of a firm can be sold for a net price of ₹ 96.50. the coupon rate of interest is 14% per annum, & the debentures will be redeemed at 5% premium on maturity. The firm's tax is 40%. Compute the AT cost of debentures.

Sol - n = 5 yrs, ~~NP~~ = price of debentures = ₹100, Net proceeds = ₹96.5

Redeemable value = ₹100 + 5% on ₹100, Interest = 14% on ₹100, = ₹14

= 105

T = 40% = 0.4

After tax = $\dfrac{I(1-t) + \dfrac{RV-NP}{n}}{\dfrac{RV+NP}{2}} = \dfrac{14(1-0.4) + \dfrac{105-96.5}{5}}{\dfrac{105+96.5}{2}}$

$= \dfrac{8.4 + \dfrac{8.5}{5}}{\dfrac{201.5}{2}} = \dfrac{\dfrac{10.1}{5}}{\dfrac{201.5}{2}} = 3.38$

$= \dfrac{14 \times 0.6 + 1.7}{100.75}$

$= \dfrac{8.4 + 1.7}{100.75} = \dfrac{10.1}{100.75} \times 100 = 1.0025$

10.025%.

Cost of prefence ~~Capital~~ Share Capital

$K_p = \dfrac{D}{NP}$ Irredeemable, $K_p = \dfrac{D + \dfrac{RV-NP}{n}}{\dfrac{RV+NP}{2}}$ Redeemable

where D = Dividend of Preference Share

A company issues 10,000 @ 10% preference shares 100 each, cost of issue is 2 per share. Calculate cost of preference capital.

a) at par (present value) b) at a premium of 10% c) at a discount of 5%

Sol Total preference capital =

a) 10,000 share × ₹100 = ₹10,00,000

$K_p = \dfrac{D}{NP}$, D = ₹10,00,000 × 10% (Preference) = 1,00,000, NP = Total preference capital = 10,00,000

less = Cost of issue = 20,000 (10000sh × 2)

9,80,000

$K_p = \dfrac{1,00,000}{₹10,00,000} \times 100 = \cancel{10\%}\ 10.20\%$.

b) at a premium of 10%

Net proceed (NP) =

Total preference capital = ₹10,00,000

Add- Premium 10L × 10% = 1,00,000

11,00,000

less- Cost of issue per share ₹ 20,000
10000 × ₹2 / share.
Np: ₹10,80,000

$Kp = \frac{D}{Np} = \frac{₹1,00,000}{10,80,000} \times 100 = 9.25\%$

c) at a discount of 5%.

Net proceeds:
Total preference Capital = ₹10,00,000
less:- discount 5% × 10L = ₹ 50,000
₹9,50,000

less:- Cost of issue ₹ 20,000
10,000 × ₹2
Np = 9,30,000

$Kp = \frac{₹1,00,000}{₹9,30,000} \times 100 = Kp = 10.75\%$

A Company's issue 10,000 @ 10% preference shares of ₹100 each redeemable after 10 yrs at a premium of 5%. Cost of issue each ₹2 per share. Calculate cost of preference capital.

Sol- Total preference capital = 10,000 × 100 = 10,00,000
+ premium 5% on 10,00,000 = 50,000
₹10,50,000 } Redeemable value.

Net proceeds - Total preference capital = 10,00,000
less- cost of issue ₹2 × 20k = 20,000
9,80,000

Dividends = 10,00,000 × 10% = 1,00,000

$Kp = \frac{D + \frac{Rv - NP}{n}}{\frac{Rv + NP}{2}} \times 100 = Kp = \frac{1,00,000 + \frac{10,50,000 - 9,80,000}{10}}{\frac{10,50,000 + 9,80,000}{2}} \times 100$

$= \frac{1,00,000 + 7000}{10,15,000} \times 100 = \frac{1,07,000}{10,15,000} \times 100 = 10.5\%$

= 10.5%

A Company issue 1,000 7% preference share of ₹100 each at premium of 10% redeemable after 5 yrs at par. Calculate cost of preference capital.

Total preference capital = 1,000 sh × 100 = ₹1,00,000.
D = ₹1,00,000 × 7% = 7,000

NPRV at par = Total preference capital = ₹1,00,000
NPRV = Total preference capital = ₹1,00,000
Add- at premium 10% on 10L = ₹10,000
₹1,10,000

D = ₹1,00,000 × 7%
= ₹7,000
Rv = ₹1,00,000
J =

$Kp \frac{D + \frac{Rv - NP}{n}}{\frac{Rv + NP}{2}} = \frac{7,000 + \frac{1,00,000 - 1,10,000}{5}}{\frac{1,00,000 + 1,10,000}{2}}$

$= \frac{7000 + (-2000)}{1,05,000} = \frac{5000}{1,05,000} = 0.04 \times 100 = 4.76\%$

Q) A Company issued 12% redeemable preference shares of ₹100 each at 5% premium redeemable after 15 yrs at 10% premium. If the floatation cost of each share is ₹2, what is the value of cost of preference share.

Sol- n = 15 yrs
D = 12% × 100 = ₹12
Rv = 100 + 10% × ₹100 = 110
Np = ₹100 + 5% × 100 - ₹2 = 103

$Kp = \frac{D + \frac{Rv - NP}{n}}{\frac{Rv + NP}{2}} = \frac{12 + \frac{110 - 103}{15}}{\frac{110 + 103}{2}} = \frac{12 + 0.4}{106.5} = \frac{12.4}{106.5} = 0.116 \times 100$

= 11.64

Q A Company issues 1000 Equity shares for ₹100 each at a premium of 10%. The Company has been paying 20% dividend to Equity Shareholders for the past 5 yrs & Expect the same in future also. Calculate the cost of equity capital will it make any difference if the market price of Equity shares is ₹160.

Sol $Ke = \frac{D}{Np}$

Here, Dividend (D) = ₹100 × 20%
= ₹20

$N_p = ₹100 + ₹100 \times 10\%$
$= ₹100 + ₹10 = ₹110$

$K_e = \frac{D}{N_p} = \frac{20}{110} \times 100 = 18.18\%$

$= \frac{20}{2160} = 0.125 \times 100 = 12.5\%$

Q) A Company's after tax cost of capital of the specific sources of finance is as under:

Sources	Book value ₹	Market value	After tax cost (%)
Equity	16,00,000	32,00,000	20
Preference shares	4,00,000	4,80,000	15
Retained Earning	8,00,000	–	18
Debentures	12,00,000	13,20,000	8
	40,00,000	50,00,000	

Compute WACC based on book value & market value weights separately.

Sol- Computation of weighted Average cost of capital ~~Source~~

Source	Bookvalue	Proportion (a)	cost (b)	WACC(axb)
Equity	16,00,000	$\frac{16}{40} = 0.4$	20	8
Preference shares	4,00,000	$\frac{4}{40} = 0.1$	15	1.5
Retained Earnings	8,00,000	$\frac{8}{40} = 0.2$	18	3.6
Debentures	12,00,000	$\frac{12}{40} = 0.3$	8	2.4
	40,00,000			15.5

Source	Market value	proportion (a)	cost (b)	WACC (axb)
Equity	32,00,000	$\frac{32}{50} = 0.64$	20	12.8
P.S	4,80,000	$\frac{48}{50} = 0.096$	15	1.44
R.E	–	–	18	–
Deben	13,20,000	$\frac{13.2}{50} = 0.264$	8	2.112
	50,00,000			16.352